

An Acoustic Investigation of Mid Front Vowel Harmony in Assamese

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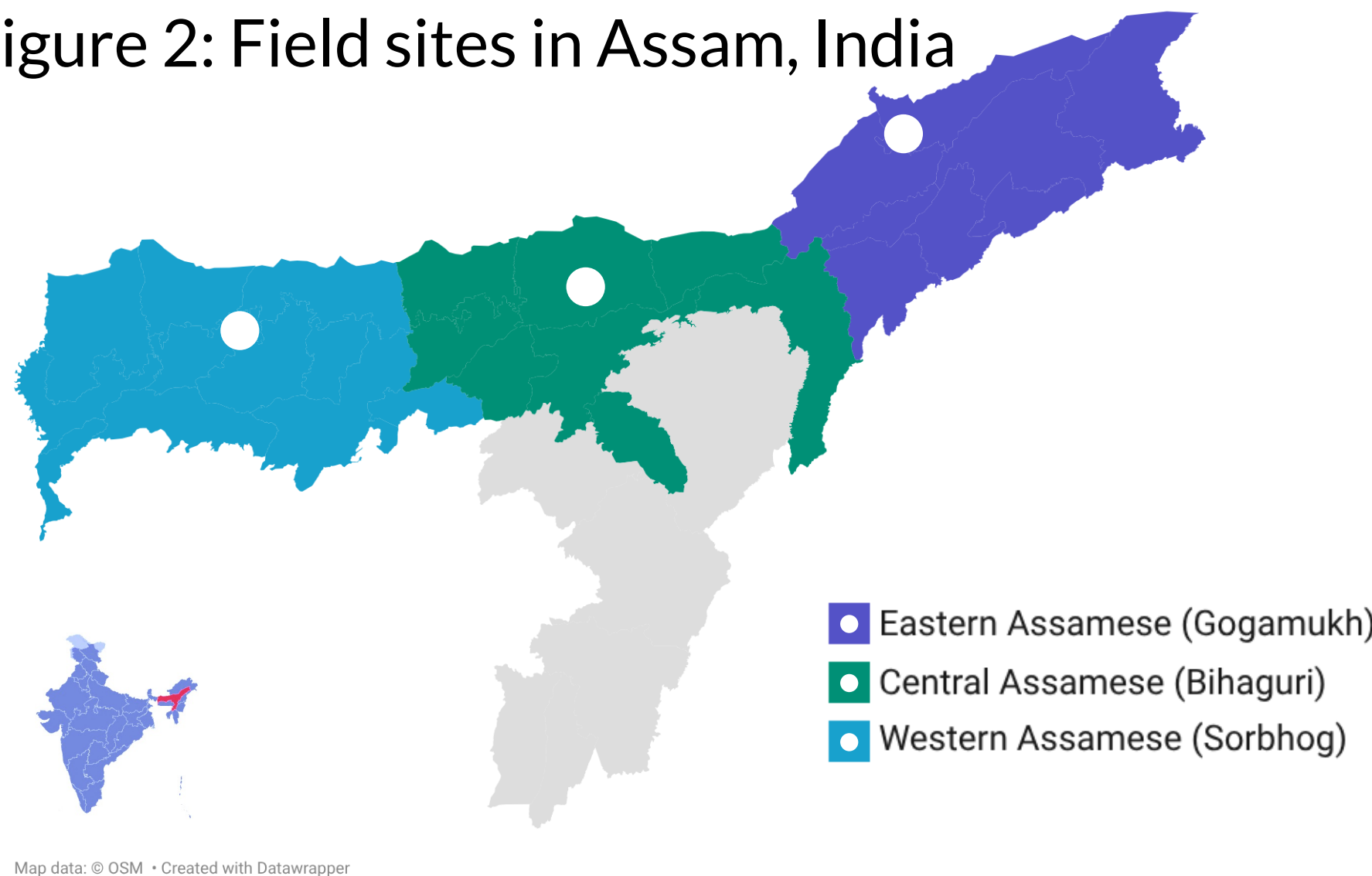
1. Introduction

- Vowel harmony: Vowels in adjacent syllables become more similar.^[1]
 - Lena Asturian: mid vowels /e, o/ raise to [i, u] before suffixes containing a close vowel.^[2]
 - Lopit: [+ATR] vowels trigger vowel harmony, causing [-ATR] vowels to surface as [+ATR] vowels.^[3]
- Most research on vowel harmony focuses on phonological patterns.
- Less is known about phonetic aspects.^[4]
- Assamese:
 - Indo-Aryan language, spoken in Assam, India.
 - /ɛ/ is realized as [e] in /CɛCV/ words, where V is a close vowel like /i/. For example: [beta] 'son' but [beti] 'daughter' ^[5]
 - Competing accounts of the harmony process: height mutation, neutralisation, [ATR].^{[5], [6]}
 - Data mostly from Standard/Eastern Assamese.^{[5], [6]} Ultra-sound imaging shows interspeaker variation.^[7]
 - Vowels show phonetic differences in regional dialects.^[8] Little is known about harmonic patterns in regional dialects.

3. Methods

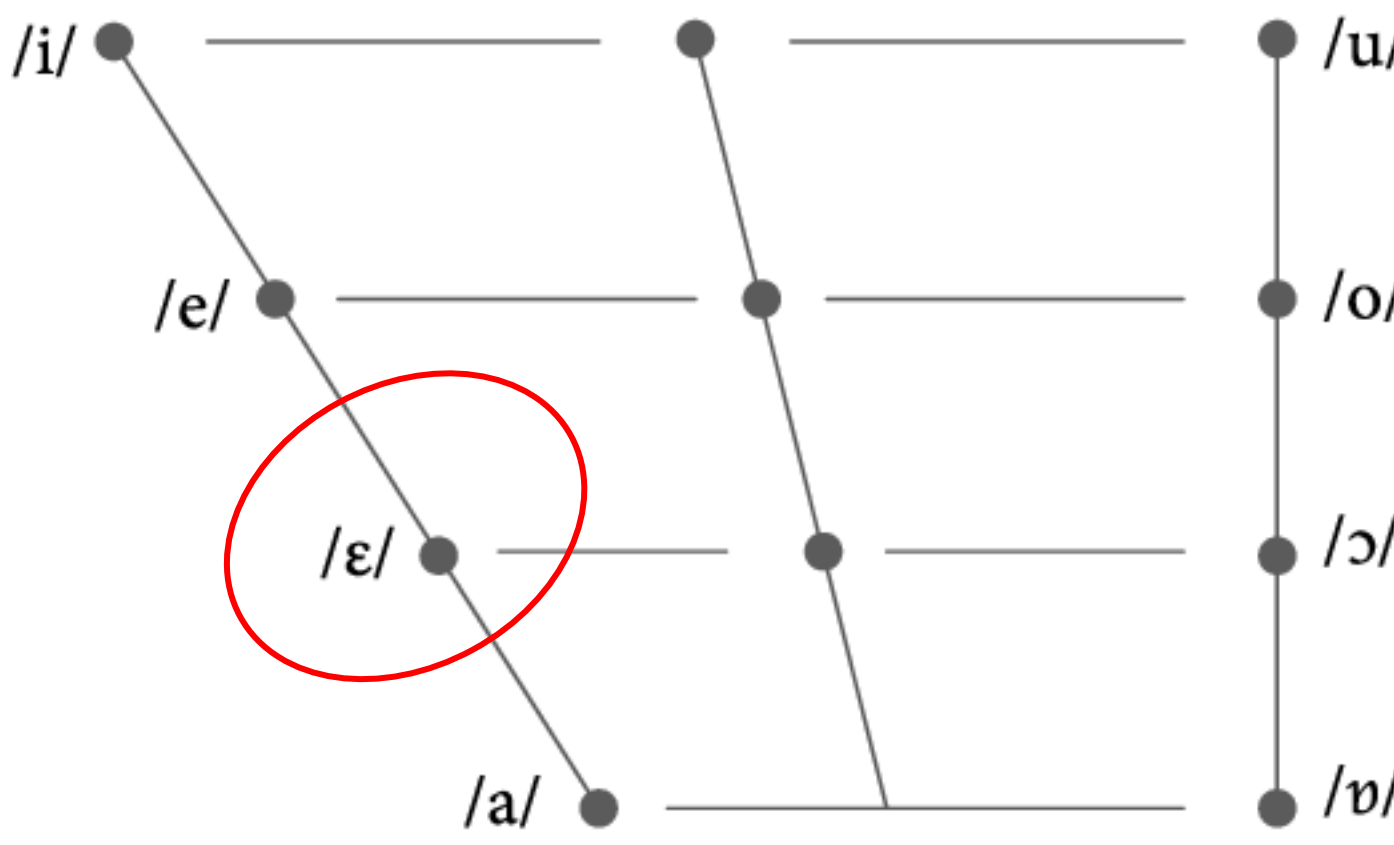
- 18 native Assamese speakers (50-65 years) from the Corpus of Contemporary Assamese Speech (COCAS).^[9]
- Speakers from three dialect regions: Western (Sorbhog), Central (Bihaguri), and Eastern Assamese (Gogamukh), 3 male and 3 female speakers per dialect.

Figure 2: Field sites in Assam, India



- 212 wordlist and 159 connected speech tokens.
- Wordlist data:
 - Each word produced three times in a carrier sentence as: *moi X buli kolo* ('I said X').
 - Target words were: /bɛki/, /sɛki/ /bɛka/, and /sɛka/
- Connected speech data: One-to-one interview; ≥30-minute audio recordings.
- Annotation and transcription in ELAN.^[10] Vowel boundary corrections in Praat.^[11]
- F1 and F2 are estimated at vowel midpoint and normalized using the Nearey1 method.^[12]
- Statistical Analysis in R: *lme4*, *emmeans*, *ggplot2*^{[13], [14], [15], [16]}

Figure 1: Vowel inventory of Standard Assamese following Goswami (1966)^[5]



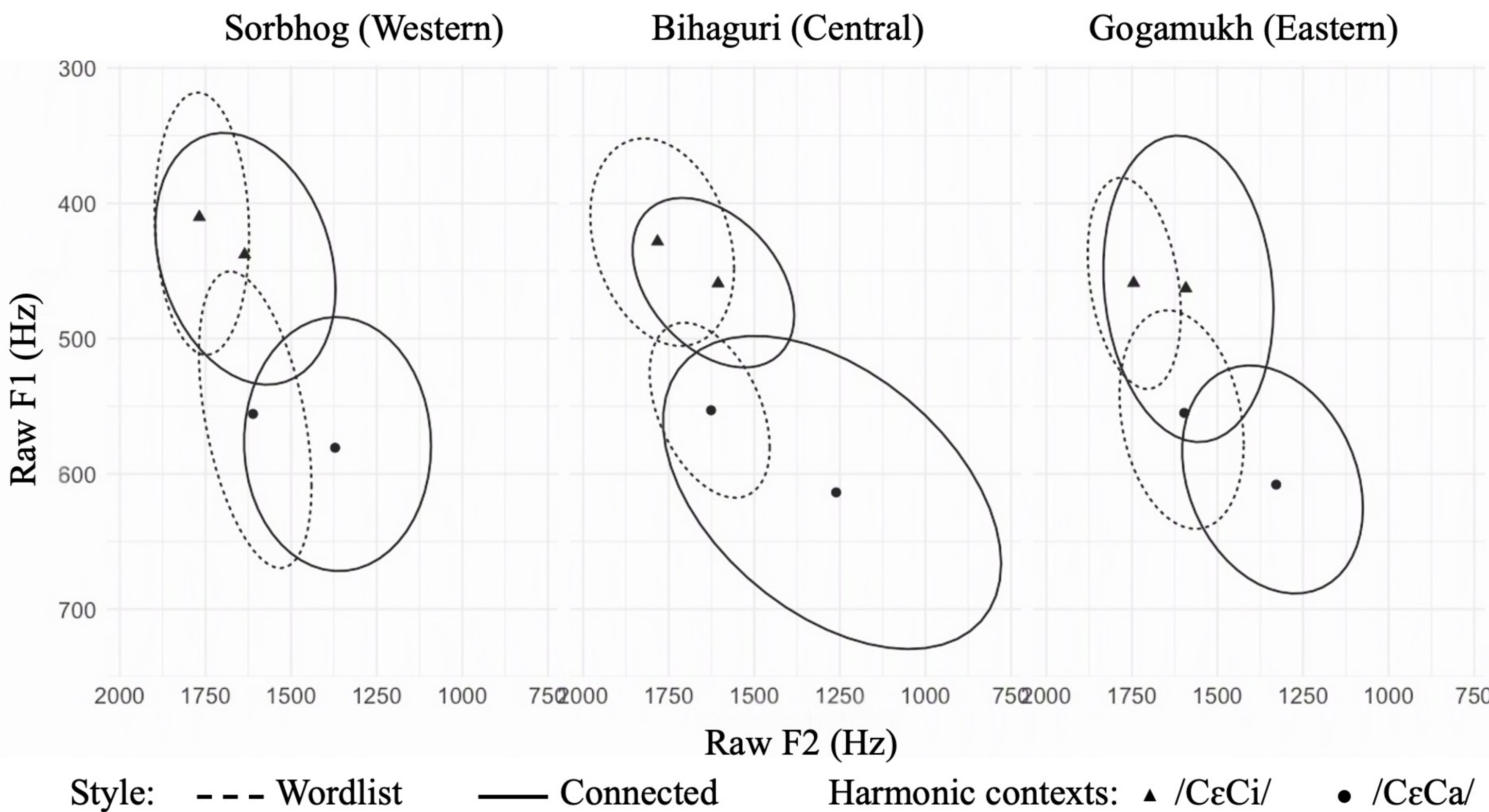
2. Research aims

- Examine the phonetic evidence for vowel harmony based on acoustic realization of /ɛ/ in harmonic (/CɛCi/) and non-harmonic (/CɛCa/) contexts.
- Assess the effects of dialect, gender, duration, and speech style.

4. Results

- /ɛ/ in /CɛCi/ words showed lower F1 and higher F2 across all dialects and styles than /CɛCa/ words.
- F1 showed a significant interaction between harmonic context, style, and dialect region, suggesting that the magnitude of vowel raising varies across dialect-style combinations.
- F2 showed a more stable pattern, with harmonic-context effects remaining largely consistent across dialects and speech styles.
- Speech style affected vowel production, with wordlist speech displaying more peripheral vowel targets than connected speech.
- Duration and gender affected acoustic variation but did not explain the harmony patterns.

Figure 3: Raw F1& F2 by context, dialect & style at 95% CI



5. Discussion

- The results support previous descriptions of Assamese mid-front vowel harmony.
- Harmony occurs across all three dialect regions, not only the Standard variety.
- Both vowel height (F1) and frontness (F2) distinguish harmonic from non-harmonic contexts, but the stronger context-dependent patterning in F1 suggests that vowel height plays the more prominent role in the harmony pattern.
- Harmony persists in both controlled (wordlist) and conversational (connected) speech.
- Dialect and speech-style differences affect how harmony is phonetically realized.
- The findings open avenues for investigating how linguistic and social factors shape vowel harmony and the vowel system more generally in Assamese.

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